King's College London

This paper is part of an examination of the College counting towards the award of a degree. Examinations are governed by the College Regulations under the authority of the Academic Board.

Examination Period	August (Period 3)
Module Code	6CCS3HCI
Module Title	Human-Computer Interaction

Format of Examination	Written questions
Start time	11am BST (GMT +1)
Time Allowed	Two hours
Instructions	You are permitted to access any materials you wish, but this is not mandated and is not expected. You may use a calculator if you find this helpful.
Rubric	ANSWER ALL QUESTIONS

The rubric for this paper must be followed and extra answers should not be submitted. For answers that are handwritten, write with blue/black ink on light coloured paper. Include the Module code, question number and student number on every page to be submitted. For answers that are typed, use the template provided.

Submission Deadline	
Submission Process	Work must be submitted to the **level 6** Informatics Assessments KEATS page.

Your work must be submitted as a PDF file. If you have prepared some answers on computer, and some on paper (which have then been digitised), you may upload at most two PDF files – one for computer-prepared answers, one for digitised answers. Do not duplicate answers across the two PDFs – if you do this, the computer-prepared answer will be taken. You should check that your work displays correctly after it has been uploaded.

ACADEMIC HONESTY AND INTEGRITY: Students at King's are part of an academic community that values trust, fairness and respect and actively encourages students to act with honesty and integrity. It is a College policy that students take responsibility for their work and comply with the universitys standards and requirements.
By submitting this assignment, I confirm that this work is entirely my own, or is the work of an assigned or permitted group, of which I am a member, with exception to any content where the works of others have been acknowledged with appropriate referencing. I also confirm that I have read and understood the College's Academic Honesty & Integrity Policy:
`https://www.kcl.ac.uk/governancezone/assessment/academic-honesty-integrity`
Misconduct regulations remain in place during this period and students can familiarise themselves with the procedures on the College website at
`https://www.kcl.ac.uk/campuslife/acservices/academic-regulations/assets-20-21/g27.pdf`

1. Foundational Concepts

a. In a few sentences, describe some of the affordances and signifiers in Figure 1.

Figure 1: iOS-style slider for controlling playback speed on a touch-screen device.

[6 marks]

b. Considering Figure 1 again, describe how its design supports users in overcoming potential gulfs of evaluation and execution.

Important: there are many different correct answers here. We want to understand how you recall and apply the terminology.

[6 marks]

c. When providing an authentication code, why might a designer express '485927492752' as '485927 492752'. What principle are they employing and why?

[4 marks]

d. Name three of Neilsen's usability heuristics and how they might help inform the design of technology. If it helps, you can describe in terms of a particular existing app (e.g. Deliveroo, Spotify, Amazon).

[9 marks]

2. **Empathise**

a. You have been hired by RunningTours, a platform for people wanting to explore a new city while running together. You aim to improve the user experience for these runners by better understanding their needs.

You are to interview 15 future users from your population. Assume you have already ethics, consent and have introduced the interviewee to the topic.

i. Briefly describe your recruitment process and target demographic.

ii. Draft an interview guide you could use, including 8-10 key questions you would ask.

iii. Outline the key areas these questions will be covering – i.e., what aspects of the users' experience are you trying to learn more about and why.

[17 marks]

b. Sketch an empathy map diagram and fill it out with 3 observations for each quadrant that you could imagine emerging from the interviews. Do not worry about what exactly your imagined observations say; we are trying to see how you are able to think about the aims of the empathy map process, rather than how well you imagine what 'real' users would say.

[8 marks]

3. Prototype

You are part of a team developing an app which allows music tutors to offer online music classes via videoconference. The results of your emphasise phase suggest it is important for students to see video of their mistakes. Your ideate phase suggests the 'SoundCheck' feature, which allows tutors to highlight mistakes in performance while in the call, and play the relevant footage to the student so that they can offer improvements.

a. For your first prototype, your manager has asked you to make a vertical prototype. What does this mean, and why is this a good idea?

[5 marks]

b. Describe a Wizard of Oz prototype to test the minimum viable product. You may show simple sketches if you like.

[9 marks]

c. Describe the advantages of low and high-resolution prototyping approaches for this context.

[6 marks]

4. Evaluate

Your local shop started using BringItToMe, an app that coordinates food delivery. They want to know if this app gives a better experience to their customers than their previous WhatsApp delivery number.

a. Outline what would be useful evaluation metrics for your system.

i. Identify 3 qualitative or quantitative indicators that you could use to show your prototype is useful compared to the previous service.

ii. Explain why you selected these indicators, and how they might be measured.

[15 marks]

b. Outline two evaluation approaches you could use to answer the question in the context of this design problem. One should be more 'in the wild', and the other should be more controlled.

i. Describe the rationale for each choice.

ii. What questions would **not** be answered by each approach?

[15 marks]